GG-C-00225a (DSA-DM)
21 April 1966
Interim Revision of
Fed. Spec. GG-C-225
1 July 1958

INTERIM FEDERAL SPECIFICATION
CHEST, X-RAY FILM PROTECTIVE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal Specification as a valid exception to Federal Specification GG-C-225, dated 1 July 1958.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers X-ray film protective chests designed to provide storage of unexposed x-ray films, or cassettes and films.

1.2 Classification.

1.2.1 Types. X-ray film protective chests shall be of the following types:

Type I. 1/16 inch lead lining, with casters.
Type II. 1/32 inch lead lining, without casters.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND OTHER PUBLICATIONS

2.1 Specifications, standards and drawings. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-C-82	Casters, Swivel, Chair.
QQ-L-201	Lead Sheet.
QQ-S-698	Steel, Sheet and Strip, Low-Carbon.
TT-E-527	Enamel, Alkyd, Lusterless.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

FSC 6525

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

STANDARDS

Federal

FED-STD-595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center
5012 Chest, X-Ray Film Protective, 1/32 Inch Lead Lining.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, MO, Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, DC.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3.1 Preproduction sample. Unless otherwise specified (see 6.2), prior to the start of production of the contract quantity, one preproduction sample of the x-ray film protective chest shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with

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the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Material.

3.2.1 Steel. All steel sheet, strip, and plate shall conform to QQ-S-698 for Skin-passed temper-CR soft No. 4 temper, No. 1 (dull) finish.

3.2.2 Lead. Lead shall conform to QQ-L-201 for grade C or D.

3.3 Construction. Chests shall be x-rayproof and lightproof. All lead seams shall be soldered or burned, or shall overlap in such a way that x-rays will be excluded, regardless of position of angle of x-ray source. The inside dimensions of the chests shall be as shown in table I.

Table I. Inside dimensions (inches)*

Type	Depth	Length	Width
I	16 1/2	19	8
II	15 3/4	18	11 1/2

*Tolerance on all dimensions shall be ±1/8 inch.

All parts and assembly of type II chest shall be as shown on Defense Personnel Support Center Drawing 5012. The chests shall consist of an inner shell, a lead barrier, an outer shell, and a cover.

3.3.1 Inner shell. The inner shell shall be sheet steel as specified in 3.2.1 and as shown in Table II. Sections of shell shall be welded together (see 3.6.1).

3.3.2 Lead barrier. The lead barrier shall be sheet lead as specified in 3.2.2 and as shown in Table II. The inner shell shall be completely covered on end, sides, and bottom with sheet lead. The sides and ends shall be covered with a single sheet, with one vertical seam. The bottom shall be a single sheet of lead with edges up to overlap the joint between the bottom, sides, and ends. Lead barrier shall be completely free of pinholes. Seams and joints shall have protection equivalent to the body material.

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3.3.3 Outer shell. The outer shell shall be sheet steel as specified in 3.2.1 and as shown in Table II. Sections of shell shall be welded together (see 3.6.1).

3.3.4 Casters. Outer shell for type I shall have four casters mounted on the bottom of the chest. The casters shall conform to the requirements of FF-C-82, for type I, style a, material 2 or 3. A flattop caster may be substituted, providing other requirements are met. The casters shall be so placed that when turned to the worst possible position, and with the lid open, the empty chest shall not be unstable. The casters may be welded in place, or bolted. Sheet metal or self-tapping screws of any type shall not be acceptable. If bolts are used, lockwashers and nuts shall be used unless the thickness of the threaded stock is at least as great as the outside diameter of the bolts. Bolts used shall not be smaller than size 8-32, and there shall be at least three for each caster. Bolts and lead lining shall be so placed that the lead cannot be punctured.

3.3.5 Cover. The cover shall consist of an inner and outer surface with lead sandwiched between. Both inner and outer surfaces shall be sheet steel as specified in 3.2.1 and as shown in Table II. The surfaces of the cover shall be covered with 1/16 inch thick lead sheet for type I chests and 1/32 inch thick lead sheet for type II chests.

3.3.5.1 Type I. Cover shall be mounted to the body of the chest by means of a continuous hinge or hinges of the 5-knuckle type, 2 1/2 inch size. The hinges may be bolted or spot-welded in position. Sheet metal or self-tapping screws of any type are not acceptable. The cover shall be held open at an angle of approximately 120° ±5° by means of a check. A drawer pull shall be welded or bolted to the cover.

3.3.5.2 Type II. Cover-check mechanism shall be provided with a spring check mechanism of design as shown on drawing. Tension on springs shall be adjusted to result in a closing force of a value sufficient only to close cover. Mechanism shall also hold cover in vertical position when chest is open.

Table II. Thickness of material*

Type (See 1.2)	Inner shell of sheet steel (see 3.3.1)	Lead barrier (see 3.3.2)	Outer shell of sheet steel (see 3.3.3)	Cover (See 3.3.5)
I	24 gage (0.025 inch)	1/16 inch sheet	20 gage (0.037 inch)	Inner Surface, 24 gage (0.025 inch) Outer surface, 20 gage (0.037 inch)

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II	20 gage (0.037 inch)	1/32 inch sheet	18 gage (0.050 inch)	Inner surface, 20 gage (0.037 inch) Outer surface, 18 gage (0.050 inch)

*Gages are United States Standard Revised (U.S.S. Rev.).

3.4 Finish. All surfaces, including inside and outside exposed surfaces, and the unexposed surfaces inside the walls of the chest, shall be degreased and coated with a primer coat as specified. The inside and outside exposed surfaces shall be coated with acid-, alkali-, and solvent-resisting enamel, as specified. All surfaces to be enameled shall be thoroughly cleaned and chemically degreased. After degreasing, a primer coat shall be applied. The primer shall be zinc-chromate conforming to MIL-P-8585. The primer coat shall form a grounding bond and base for the finished enamel.

3.4.1 Enamel.

3.4.1.1 Type I. The primer coat shall be followed by one coat of enamel. Each coat shall be sprayed on and baked in high-temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, cracking, and chipping. The color of the final finish shall be the manufacturer's standard gray color.

3.4.1.2 Type II. The primer coat shall be followed by one coat of lusterless enamel conforming to TT-E-527. Each coat shall be sprayed on and baked in high temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, cracking, and chipping. The color of the final finish shall be lusterless green conforming to color X 34087, as specified in FED-STD-595, (see 6.1).

3.5 Identification marking. Each chest shall have a metal nameplate permanently marked with the following information, and located to permit reading without difficulty:

Manufacturer's name or registered trade mark.
Manufacturer's type and model number.
Serial number.
Contract or purchase order number.
Federal stock number.
Letters U.S.

Marking secured by adhesives shall not be used.

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3.6 Workmanship. Workmanship shall be first class throughout. The X-ray film protective chests shall be free from defects which detract from their appearance or which may impair their serviceability.

3.6.1 Welding. Seam and fillet welds shall be continuous, even and smooth. Spot welds shall be not more than 2 inch centers, and shall be accomplished so as to insure sound welds of sufficient strength and completeness to tear the spot out of the parent metal when pulled apart. All exposed seam and fillet welds shall be ground smooth to eliminate traces of weld marks, and polished to blend into adjacent surfaces. All exposed spot welds shall be ground smooth, or filled and sanded smooth so that they will not be visible after the enamel finish is applied.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling for examination and testing shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The X-ray film protective chests shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One chest, of the type specified, constitutes one unit of issue.

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5.1.2 Level A.

5.1.2.1 Unit preparation. All exterior finished surfaces shall be wrapped with kraft paper and cushioned with corrugated fiberboard or other suitable material. Wrapping and cushioning shall be adequately secured. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be overpacked in an exterior container designed for a type 2 or type 3 load, as applicable, and constructed in accordance with PPP-B-585, style 3 for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table III or table IV as applicable. Type I chests shall be packed in such a manner that the weight of the unit shall not rest on the casters.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load or a type 3 load as applicable and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I. Type I chest shall be packed as specified in 5.2.1.1 for caster protection.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Availability of color chips. The individual 3-by 5-inch chips of the color referenced in 3.4.1.2 are available from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S.W., Washington 25, D.C.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).
(c) Where preproduction sample shall be sent (see 3.1).
(d) When preproduction sample is not required (see 3.1).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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